

Predictive models of air temperatures inside a naturally ventilated vehicle transporting weaner pigs

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ABSTRACT

Maintaining proper interior thermal condition during transportation is vital for animal welfare and sustainability of livestock supply chain. This study investigated the air temperatures inside a multi-deck naturally ventilated vehicle when transporting weaner pigs under warmer weather condition by using computational fluid dynamics (CFD). Predictive models of interior air temperatures were developed by using response surface methodology (RSM) and gradient boosting machine (GBM) with the inputs of exterior air temperature, vehicle speed, wind speed, incident wind angle and opening height of shutter based on the dataset generated from CFD simulations and validated as well. The results showed that predictive models developed by RSM were sufficient for predicting the interior air temperatures of moving naturally ventilated livestock vehicle, and GBM could improve the prediction accuracy moderately. RSM models indicated that the interior temperatures increased linearly with the increase in exterior air temperature, opening height and wind speed while insensitive to vehicle speed. GBM model indicated that the plane-average air temperature of front compartments was 2.2 °C higher than those of the other two compartments at the same deck, and the air temperature increased slightly from the bottom to the upper deck. High spatial variations in air temperature were observed inside the moving livestock vehicle, which poses a challenge on monitoring interior air temperatures. The developed models are expected to predict the interior air temperatures and provide suggestion on regulating ventilation systems in advance. Further study could be conducted to investigate the optimum control of opening for improving the natural ventilation potential.

1. Introduction

Pigs are vulnerable to heat stress and have less ability to withstand high temperature during transportation (Rioja-Lang et al., 2019). The number of in-transit pig losses was reported to be increased under high interior temperature (Haley et al., 2008; Sutherland et al., 2009). High interior temperature could also increase pigs' stress (Lambooy, 1988) and induce physiological changes (Kephart et al., 2010) during transportation. Maintaining interior air temperature within thermal comfort zone of pigs during transportation is vital for animal welfare and sustainability of livestock supply chain.

Experimental studies indicated that the thermal condition inside livestock transport vehicle is affected by several factors, e.g. exterior climatic conditions, stocking density, ventilation systems, dimension of openings for natural ventilation, dimensions of vehicle, and moving status of vehicle (Randall and Patel, 1994; Christensen and Gade, 1997; Norton et al., 2013; Dobbins, 2019; Thodberg et al., 2022). During the

three journeys of transporting slaughter pigs from the Netherlands to Italy, the correlation coefficients between the exterior and interior temperatures of the three-deck trailer were 0.88, 0.81 and 0.54, respectively (Lambooy, 1988), and the correlation coefficients decreased when exterior temperature became lower. Stocking density determines the total heat production from animals and therefore has a significant impact on interior thermal conditions (Warriss, 1998). The interior air temperature was slightly lower and interior air speed was slightly higher when the stocking density was reduced during long distance transport of pigs (Gerritzen et al., 2013). The interior air temperature of vehicle transporting pigs was also affected by the moving status of vehicle. Significant increase in interior air temperature has been recorded when the vehicle was stationary (Thodberg et al., 2022) and stopped for short break (Xiong et al., 2015). The increase in air temperature was slowed down and levelled out faster by mechanical ventilation when vehicle was stationary compared to natural ventilation during the transport of pigs (Christensen and Jonsson, 2007). The

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location of sensors can affect the interpretation of interior thermal condition as well, with air temperature higher at animal level than those measured under the ceiling during the transport of feeder cattle (Goldhawk et al., 2014a). The interior thermal condition was also affected by compartment arrangements (Brown et al., 2011). Temperature variation between compartments at the same deck were noticed during transportation of pigs with higher air temperature being recorded in the front compartments (close to driver's cabin) (Christensen and Gade, 1997; Harmon et al., 2017; Foldager et al., 2022).

A large trade of weaners was transported between EU member states, but the investigation on thermal environment inside vehicles transporting weaners were still limited (EFSA Ahaw Panel et al., 2022). Our previous on-road experiments (Foldager et al., 2022) showed large horizontal variation in air temperature from the rear to front compartments inside the vehicle transporting weaner pigs when the vehicle was primarily naturally ventilated, and the sensors' location would affect the interpretation of interior temperature distribution. Due to the high cost and time consumed by on-road experiments, the influencing factors on interior air temperature of livestock transport vehicles have not been quantified and discussed thoroughly yet.

Computational fluid dynamics (CFD) has an advantage on the full control of the influencing factors and on obtaining the detailed distributions of target environmental parameters inside the entire computational domain. CFD has been employed to investigate the thermal environment of livestock building extensively (Ma et al., 2022; Mostafa et al., 2012; Norton et al., 2010; Rong et al., 2015). However, its applications on livestock transport vehicle were still limited and only a few studies can be found in literature. CFD simulations under isothermal conditions showed that the exterior flow was prone to enter from the last few openings at the rear part of the moving livestock trailer and moved forward to the front part when the interior space was not divided by partitions (Gilkeson et al., 2016). When the dual-mode transport vehicle was held still on a ferry, the mechanically ventilated deck was cooler but its temperature-humidity fields were less uniform than the naturally ventilated deck as evaluated by CFD simulations (Norton et al., 2013).

Studies showed that combining response surface methodology (RSM) and CFD was feasible and efficient for predicting the ventilation rate of naturally ventilated livestock building (Shen et al., 2013, 2012; Yi et al., 2021) and convective heat loss of pigs (Li et al., 2018a; Chen et al., 2024). Machine learning (ML) algorithms have been extensively applied to indoor thermal comfort study (Qavidel Fard et al., 2022), and they have also demonstrated their strengths in predicting air temperature inside livestock building (Arulmozhi et al., 2021; Gautam et al., 2021; Lee et al., 2022) in recent years. Compared with support vector machine (SVM) and artificial neural network (ANN), the tree-based algorithms, the most frequently used algorithms in thermal comfort studies, do not require data pre-processing and can achieve high accuracy with easy interpretation on the importance of features used in the predictions. The tree-based algorithms (random forest (RF) and gradient boosted machine (GBM)) have been successfully applied to predict the building energy consumption (Ahmad et al., 2017; Papadopoulos et al., 2018; Touzani et al., 2018) and indoor temperature (Alawadi et al., 2022). Compared with RF, GBM sequentially builds an ensemble of shallow trees in sequence with each tree learning and improving on the previous one (Boehmke and Greenwell, 2019), and therefore provide accurate and robust predictive model.

The objectives of this study were to: (1) develop the predictive model/program by using RSM and GBM to estimate interior air temperature of a pig transporting vehicle with four-deck and natural ventilation based on the dataset generated from CFD simulations and assess their prediction accuracies, respectively; and (2) analyse the influences of exterior air temperature, vehicle speed, wind speed, incident wind angle and opening height of shutters on interior air temperature based on the developed predictive models. The models are expected to be implemented into the indoor thermal control of transporting vehicles.

2. Methodology

Design of experiment (DOE) was employed to identify the cases that were going to be simulated by CFD to generate the dataset for developing the predictive models (one for incident wind angle of 0° – 90° , and the other one for incident wind angle of 90° – 180°) by RSM. For simplification, the plane-averaged temperatures at 50 cm above the compartment floors at the same column (e.g. front, middle and rear compartments) were treated as response variables, thus three predictive models were developed by adopting RSM for each range of incident wind angle, respectively. For each range of wind incident angle, the number of CFD simulation cases for developing and validating predictive models were 46 and 18, respectively.

To develop predictive models covering incident wind angle of 0° to 180° for front, middle and rear compartments, respectively, the entire CFD simulation cases (110 in total because there were 18 duplicated CFD simulation cases) were used by GBM, among which 77 cases were used for developing the predictive models and the other 33 cases were used for validation. Those three models developed by GBM were generated to identify the influence of each factor on interior air temperatures in front, middle and rear compartments, and to compare them with the values predicted by RSM models. Since the inputs of compartment location at each deck (hereafter compartment, three levels) and deck (four levels) could not be covered by DOE and RSM models, a unitary predictive model containing influencing factors of compartment and deck, exterior air temperature, vehicle speed, wind speed, incident wind angle (hereafter wind angle), and opening height of shutter (hereafter opening height) were also established by GBM for predicting the plane-averaged temperatures at 50 cm above the compartment floor of any of the twelve compartments. The unitary predictive model established by GBM employed 70 % of the CFD simulation cases for development and the rest 30 % for validation.

2.1. CFD modelling

2.1.1. Geometric models of the truck and pigs

Fig. 1b shows the schematic configurations of the investigated transport vehicle which was dimensioned following the one (see Fig. 1a) used during on-road experiment in Denmark (Foldager et al., 2022). Based on European practice, weaner pigs are transported in vehicles with four or five decks, corresponding to a deck height of approximately 70 cm and 60 cm, respectively. Considering the computational capacity, only the pigs inside lorry were modelled and the trailer was treated as sealed body in all CFD simulations except for the CFD validation cases, in which the geometric models were created as close as to the experimental conditions with reasonable simplifications. The lorry was divided into four identical decks with height of 70 cm. Each deck consists of three compartments (front, middle and rear) separated by solid partitions, as shown in Fig. 1c. The solid partitions inside each deck were identical with height of 61 cm and thickness of 3 cm. Meanwhile, gaps between the partitions and the floors and sidewalls were 7 cm and 18 cm, respectively. The openings of fans were sealed and the openings at both lateral sides were modelled as identical and followed the dimensions of openings on right side (the driver's right hand side) of the experimental vehicle. The lateral side walls and roof are not insulated, and the gap between interior and exterior walls was 5 cm. The openings height was identical in 30 cm and the bottom of the openings was positioned at 40 cm above the floor, see Fig. 1c. The interior sides of the opening were covered with window grills modelled as porous baffle. The exterior sides of openings were covered by adjustable shutters which can be adjusted from bottom to top. It is assumed that the shutters of both sides were adjusted spontaneously with the identical opening height.

The short journey with duration of approx. 8 h is allowed to have higher stocking density (Ministry of Food, Agriculture and Fisheries, 2020) and thus might be more critical for interior thermal condition in warmer period. Hence, the short journey was selected to design our

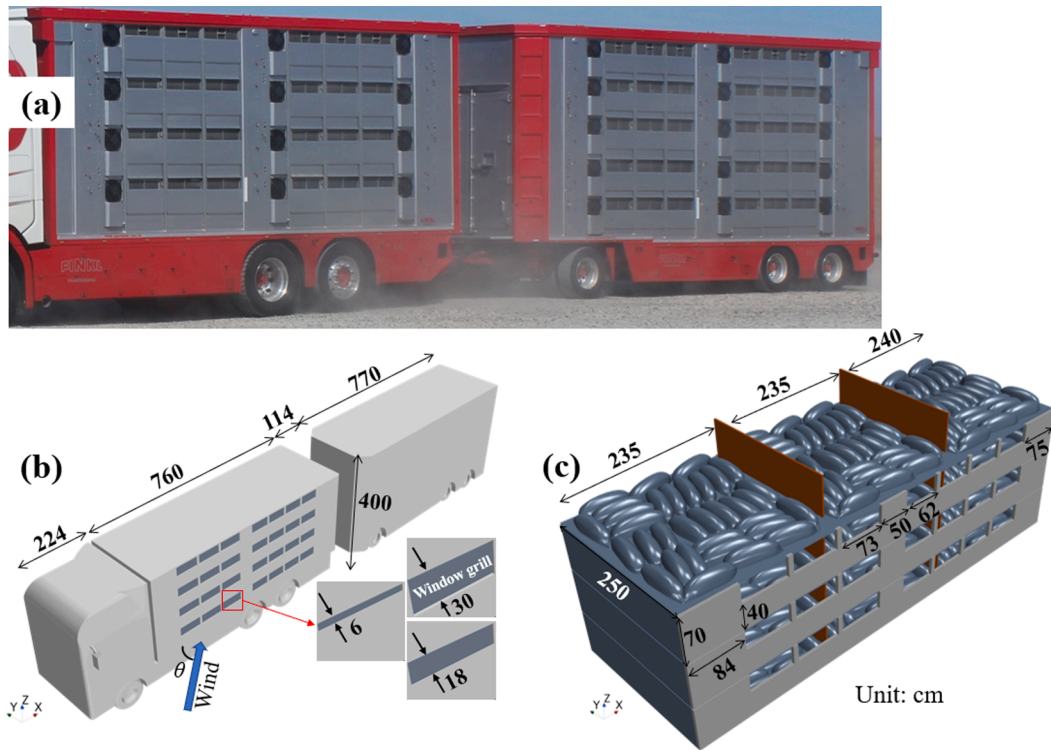


Fig. 1. (a) Exterior view of the experimental vehicle; (b) Exterior view of investigated vehicle with the opening height of shutters being illustrated in the enlarged views; (c) Geometry and dimensions of the four-decks lorry with the walls and roof of Deck-4 being hidden.

simulation scenarios. Following a typical experimental short journey (conducted on 31st August 2021), the stocking numbers in front, middle and rear compartments at each deck were set as 33, 33 and 34, respectively. The pigs were modelled as legless simplified pigs (Li et al., 2018b) and mostly lying on the floors or leaning on each other as shown by the recorded videos during the experimental short journey. The average body mass of pigs was 24 kg, with dimensions calculated following the study by Condotta et al. (2018).

2.1.2. Computational domain and mesh resolution

Cylindrical computational domain was used in CFD modelling with vehicle being placed in the center. The boundaries of inlet, top and ground in the computational domain were set as velocity inlet, symmetry plane and wall, respectively. By using cylindrical domain, wind and thermal fields under varying wind directions can be investigated without modifying and re-meshing the entire computational domain (Chen et al., 2021a, 2021b). The radius and height of the cylindrical domain were set as 160 m (40H) and 24 m (6H), respectively, where H (4 m) was the height of the vehicle.

Fig. 2 presents partial views of the mesh distribution. The mesh was generated by Polyhedral Mesher with surface control. Prism layers were employed to achieve $y^+ < 5$ at the pigs' body surface and $y^+ > 35$ at all the other surfaces. The base size was set as 2.0 m with surface growth rate of 1.3. The surface sizes of pigs, deck floors and window grills were set as 0.5 % of the base size. The surface sizes of ceilings, partitions and interior walls of the vehicle were set as 2 % of the base size. The surface sizes of exterior walls of the vehicle were set as 4 % of the base size. The surface sizes of inlet, ground and top were 5 %, 20 % and 40 % of the base size, respectively. The final grid number for all the investigated cases was around 33.0 million. Finer mesh could lead to cells number higher than 100 million, which is beyond our current computational capacity. Therefore, mesh independent study has not been conducted in this study.

2.1.3. Inflow boundary conditions

A prescribed log-law profile of atmospheric boundary layer (ABL) was defined at the inlet. The vertical profiles of velocity $U(z)$ ($\text{m}\cdot\text{s}^{-1}$), turbulent kinetic energy $k(z)$ ($\text{m}^2\cdot\text{s}^{-2}$) and the dissipation rate of the

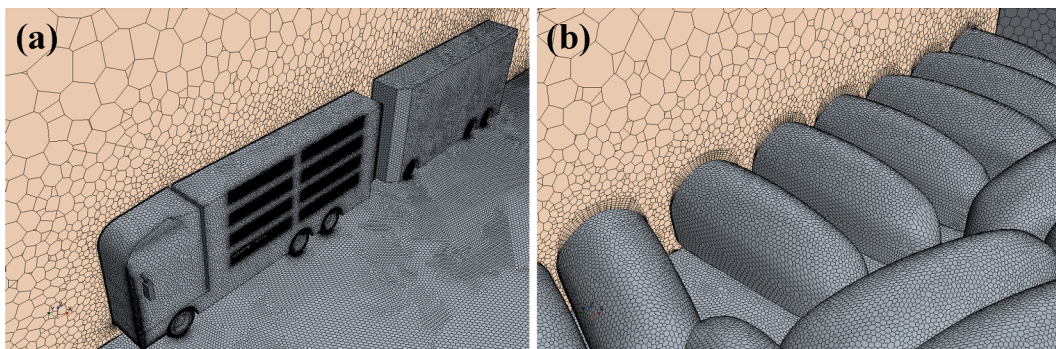


Fig. 2. Mesh distributions near (a) the vehicle and the ground and (b) the simplified pigs. The yellow panels were the central vertical plane. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

turbulent kinetic energy $\varepsilon(z)$ (m^2s^{-3}) were denoted by Eqs. (1) to (3). No-slip condition was used at all solid surfaces. The ground was set as rough wall to reproduce the horizontally homogeneous atmospheric boundary layer. The roughness constant C_s was set as 4.0. The relationship between the roughness length and roughness height (k_s) was shown in Eq. (5).

$$U(z) = \frac{u_{ABL}^*}{k_v} \ln\left(\frac{z+z_0}{z_0}\right) \quad (1)$$

$$k(z) = \frac{u_{ABL}^{*2}}{\sqrt{C_\mu}} \quad (2)$$

$$\varepsilon(z) = \frac{u_{ABL}^{*3}}{k_v(z+z_0)} \quad (3)$$

$$u_{ABL}^* = \frac{U_{ref} k_v}{\ln\left(\frac{z_{ref}+z_0}{z_0}\right)} \quad (4)$$

$$k_s = \frac{9.793z_0}{C_s} \quad (5)$$

where the atmospheric boundary layer friction velocity (u_{ABL}^*) was computed with reference wind velocity (U_{ref}) at reference height ($z_{ref} = 5$ m). The design variable wind speeds (W_{sp}) were adopted as U_{ref} . C_μ is a constant (0.09), and k_v is von Karman's constant (0.41). z_0 is the roughness height (0.1 m).

The vehicle was modelled as stationary. To mimic the movement of vehicle, tangential velocities (local rotation rate) were applied to lorry wheels and trailer wheels with wall relative rotations transferred from vehicle speed and wheels' diameters. Tangential velocity (vector) was applied to the ground with relative velocity equalled to vehicle speed (V_{sp}) but moving in opposite direction. The vehicle's moving direction was set as $-x$ (see Fig. 1b). When the vehicle moved with speed of V_{sp} , the resultant vertical profiles of time-averaged velocity components ($\bar{u}(z)$, $\bar{v}(z)$ and $\bar{w}(z)$) at the inlet were the three components of addition of vehicle velocity vector and wind velocity vector (Abdessalem et al., 2012). With wind blowing the vehicle with wind angle of θ (as Fig. 1b shows), the $\bar{u}(z)$, $\bar{v}(z)$ and $\bar{w}(z)$ were expressed as follow:

$$\bar{u}(z) = V_{sp} + U(z)\cos\theta \quad (6)$$

$$\bar{v}(z) = U(z)\sin\theta \quad (7)$$

$$\bar{w}(z) = 0 \quad (8)$$

2.1.4. Thermal boundary conditions

The radiation was not modelled in current CFD simulations. Preliminary study showed that the exterior wall's temperature of the moving vehicle with relative high speed was close to the exterior air temperature under solar radiation. Therefore, the exterior air temperature was applied to the ground and the external walls of the moving vehicle. The interior walls of vehicle (excl. floors) were set as convective with heat transfer coefficient of $8.0 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$ (Jayamaha et al., 1996). Floors, which were covered by sawdust when loaded with pigs, were applied with adiabatic thermal condition. The radiation and evaporation were not modelled, and sensible heat production of pigs was modelled as that 40 % of the sensible heat flux was released to the air. The total heat production of pigs was calculated by equations suggested by International Commission of Agricultural Engineering (CIGR) (Pedersen and Sällvik, 2002) for fattening pig.

The total heat production (ϕ_{tot} , W) of fattening pig at 20°C was calculated by Eq. (9):

$$\phi_{tot} = 5.09m^{0.75} + [1-(0.47 + 0.003m)] [n \times 5.09m^{0.75} - 5.09m^{0.75}] \quad (9)$$

where m represents body mass (kg) and n represents daily feed energy

intake. For a fattening pig with body mass from 20 kg to 30 kg, n is 3.35.

The sensible heat production (ϕ_s , W) per hpu from fattening pig on partly or completely solid floor in Northern Europe is calculated by Eq. (10):

$$\phi_s = 0.62[1000 + 12 \times (20-t)] - 1.15 \times 10^{-7} \times t^6 \quad (10)$$

where one hpu corresponds to 1000 W of total heat production at 20°C , t is the ambient temperature ($^\circ\text{C}$).

2.1.5. Solver setting of CFD modelling

STAR-CCM+ (2021.3) (CD-adapco, 2021) was employed for our CFD modelling. The Realizable $k-\varepsilon$ Two-Layer model was selected to solve the turbulent quantities and the energy conservation equation was activated to solve the temperature distributions. The buoyancy force generated by the temperature difference was accounted for by using Boussinesq's approximation. The segregated solver was adopted to solve the pressure-velocity fields. The second-order upwind scheme was selected to discretize the convection terms of the partial differential equations. The stopping criteria were 10^{-4} for continuity, energy, turbulent dissipation rate, turbulent kinetic energy and velocity components (x , y and z). The minimum iteration was 4000 and the convergence was not considered to be reached until plane-average air temperature at 50 cm above the floor in each compartment changed little.

To compensate for the differences between steady-state and unsteady-state simulations, the simulations were run in twofold. Firstly, the heat flux values of pigs in each of the twelve compartments were identical and calculated with the investigated exterior air temperature. After this initial simulation converged, the heat flux values of pigs in each of the twelve compartments were updated with the plane-average temperatures at 50 cm (40 cm in the cases of CFD validation) above their floors correspondingly. The updated simulation was run at least 4000 more iterations until converged and plane-average air temperatures became constant.

2.1.6. Validation of CFD modelling

Field trials, 16 short (8 h) journeys and 16 long (23 h) journeys, were conducted from June 2021 to March 2022 to investigate the impact of deck height on indoor air temperature inside a livestock vehicle when transporting weaner pigs through Denmark (Foldager et al., 2022). The vehicle consists of a four-deck lorry and a five-deck trailer, as shown in Fig. 1a. Only the Deck 1 and Deck 3 were loaded with 20–25 kg weaner pigs during the experimental journeys. Two levels of deck heights were considered in the lorry (70 cm vs. 90 cm) and trailer (60 cm vs. 80 cm). Each deck was partitioned into three compartments with stocking density of 30–40 and 26–31 pigs per compartment for short and long journeys, respectively. The geographical position of vehicle recorded by GPS system, opening height of shutters and activation status of fans (on/off) were logged every 5 min for most journeys. The air temperatures inside each compartment were recorded at one-minute intervals. The exterior weather conditions (dry-bulb temperature, wind speed, wind direction, etc.) were retrieved from nearest weather stations based on GPS file by making API calls to OpenWeather. The vehicle speed and vehicle's direction of motion were identified based on GPS data.

The time slot from 13:00 to 14:00 during a short journey on 31st August 2021, when the temperature differences between interior and exterior were relatively stable, was chosen as the validation scenario. The detailed input parameters for validation were summarized in Table 1. Considering their thickness, the window grill (5 mm) and fan (28 mm) were modelled as porous baffle and porous region, respectively. The properties of window grill and fan were obtained from CFD sub-studies by using the dimensions measured from the experimental vehicle. The porous inertial resistance and porous viscous resistance were calculated based on the pressure drop across window grill or fan by Eqs. (11) and (12). It should be noted that the P_i and P_v of fan in Table 1 were set in yy direction, and those values for xx and zz directions were

Table 1

Geometrical dimension and boundary conditions for validation study and the properties of window grills were also used in other CFD cases.

Parameters	Values
Deck height of 1st floor	Lorry: 90 cm; Trailer: 60 cm
Deck height of 3rd floor	Lorry: 70 cm; Trailer: 80 cm
Opening height of left-shutters	Lorry: 12 cm; Trailer: 5 cm
Opening height of right-shutters	Lorry: 8 cm; Trailer: 8 cm
Vehicle speed	80.6 km·h ⁻¹
Wind speed	2.0 m·s ⁻¹
Exterior temperature	23.7 °C
Window grill(Porous baffle)	Porosity: 0.86 α : 0.385 β : 0.114 m·s ⁻¹
Fan(Porous region)	Porosity: 0.16 P_i : 465.54 kg·m ⁻⁴ P_v : 11.89 kg·m ⁻³ ·s ⁻¹

Table 2

Table of design variables and design spaces.

Design variables	Symbol	Unit	Design space
Opening height	H	cm	6–30
Vehicle speed	V_{sp}	km·h ⁻¹	40–80
Exterior temperature	T_{ext}	°C	14–26
Wind speed	W_{sp}	m·s ⁻¹	0–10
Wind angle	θ	°	0–90; 90–180

set to 10000. The CFD settings in validation case followed the above-mentioned settings in subsections 2.1.2 to 2.1.5. The Realizable $k-\epsilon$ Two-Layer model (RKE2L) and Standard $k-\epsilon$ Two-Layer model (SKE2L) were both employed in CFD simulations for the validation. Details of obtaining the pressure drops for calculating the inertial resistance and viscous resistance in porous medium modelling of wind grill and fan can be found in the [Appendix](#).

$$\frac{\Delta P}{L} = -(P_i|v| + P_v)v \quad (11)$$

$$\Delta P = -\rho(\alpha|v| + \beta)v \quad (12)$$

where, ΔP (Pa) is the pressure drop across the porous region/baffle; v (m·s⁻¹) is the superficial velocity through the porous region/baffle; ρ (kg·m⁻³) is the fluid density; P_i (kg·m⁻⁴) is the porous inertial resistance of porous region; P_v (kg·m⁻³·s⁻¹) is the porous viscous resistance of porous region; α is the porous inertial resistance of porous baffle; β (m·s⁻¹) is the porous viscous resistance of porous baffle.

2.2. Design of CFD simulation scenarios by employing DOE

The design spaces of the investigated variables in DOE were

summarized in [Table 2](#). The opening height of shutters in the lorry during the experimental short journeys under warmer weather changed from 6 cm to 30 cm (see [Fig. 1b](#)) and was thus selected for DOE. The highest vehicle speed during the experiment trials was approx. 80 km·h⁻¹ and the speed limit is 40 km·h⁻¹ in certain built-up areas, the range of vehicle speed was set from 40 km·h⁻¹ to 80 km·h⁻¹ for DOE. The exterior air temperature changed from 14 °C to 26 °C per the recorded lowest and highest exterior temperatures between 09:00 to 16:00 during the short journeys under warmer weather, and that range was also observed in TMY (Typical Meteorological Year) weather data file of Copenhagen. The wind speed was investigated at a range of 0 m·s⁻¹ to 10 m·s⁻¹, which matched both weather data extracted from nearest weather station during the short journeys under warmer weather and TMY weather data of Copenhagen. As the openings at both sides were set as identical, it is sufficient to investigate wind angle in the range of 0°–90°. The wind angle was further divided into two design spaces, 0°–90° and 90°–180°, for developing and validating RSM models.

The design of experiment was conducted in R ([R Core Team, 2021](#)) with 'rsm' package. The Box-Behnken design (BBD) method was adopted in DOE with each design variable divided into three equally spaced levels: low level, medium level, and high level, which represents lower value, medium value, and upper value of their respective design spaces. The 5-factor BBD has 2 blocks, and each block has 3 center points. The designed cases for developing RSM models with wind angle ranging from 0° to 90° and 90° to 180° were listed in [Table SM.1](#) and [Table SM.2](#) of the [supplementary material](#), respectively. Totally 92 cases were designed with 18 duplicated cases. Meanwhile, 36 more cases were also designed for validation, as summarized in [Table SM.3](#) and [Table SM.4](#) of the [supplementary material](#). These validation cases were designed with the middle values of the lower and medium values of design space and middle values of the medium and upper values of design space, i.e. 12 cm and 24 cm for opening height, 50 km·h⁻¹ and 70 km·h⁻¹ for vehicle speed, 17 °C and 23 °C for exterior temperature, 2.5 m·s⁻¹ and 7.5 m·s⁻¹ for wind speed, 22.5° and 67.5° for wind direction (0°–90°), 112.5° and 157.5° for wind direction (90°–180°). By doing so, the transitions of polynomial curves between design spaces could be better validated, and the entire dataset is more adequate for the development of GBM models. The sample function in R was employed to randomly select 18 cases from the 32 possible combinations of the abovementioned parameters for wind direction of 0° to 90° and 90° to 180°, respectively.

2.3. Development and validation of predictive models

2.3.1. Response surface methodology (RSM)

RSM is a collection of mathematical and statistical techniques used to model and analyze the functional relationship between a target response and multi independent variables ([Khuri and Mukhopadhyay, 2010](#)). The RSM, associated with the DOE, is used to analyze the quadratic response surfaces and develop second-order polynomial models, as expressed in Eq. (13).

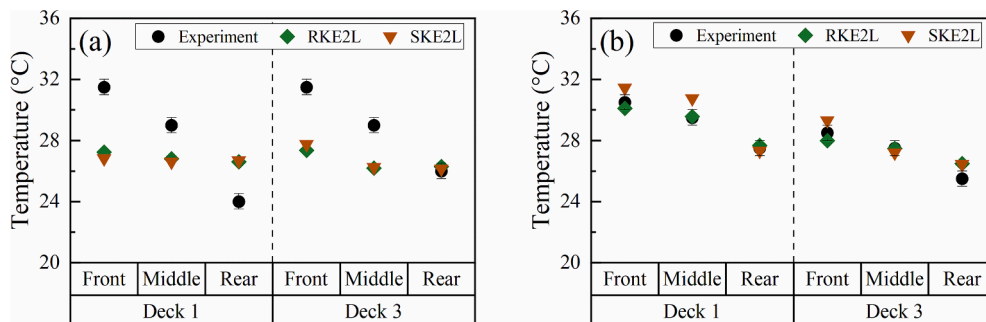


Fig. 3. CFD validation results of air temperatures inside the (a) lorry and (b) trailer.

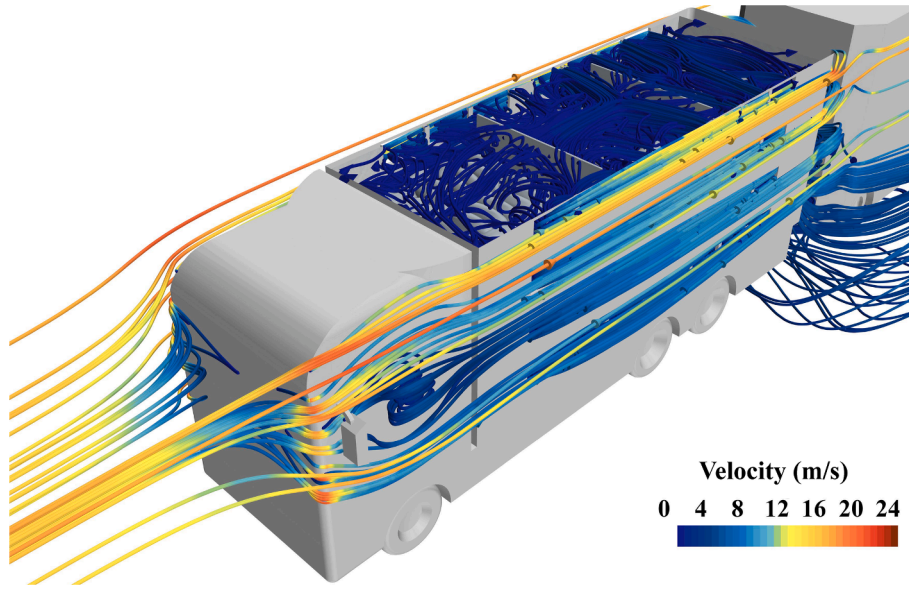


Fig. 4. Streamline in the case of $V_{sp} = 60 \text{ km}\cdot\text{h}^{-1}$, $H = 30 \text{ cm}$, $T_{ext} = 20 \text{ }^{\circ}\text{C}$, $W_{sp} = 5 \text{ m}\cdot\text{s}^{-1}$ and $\theta = 0^{\circ}$ with roof being hidden.

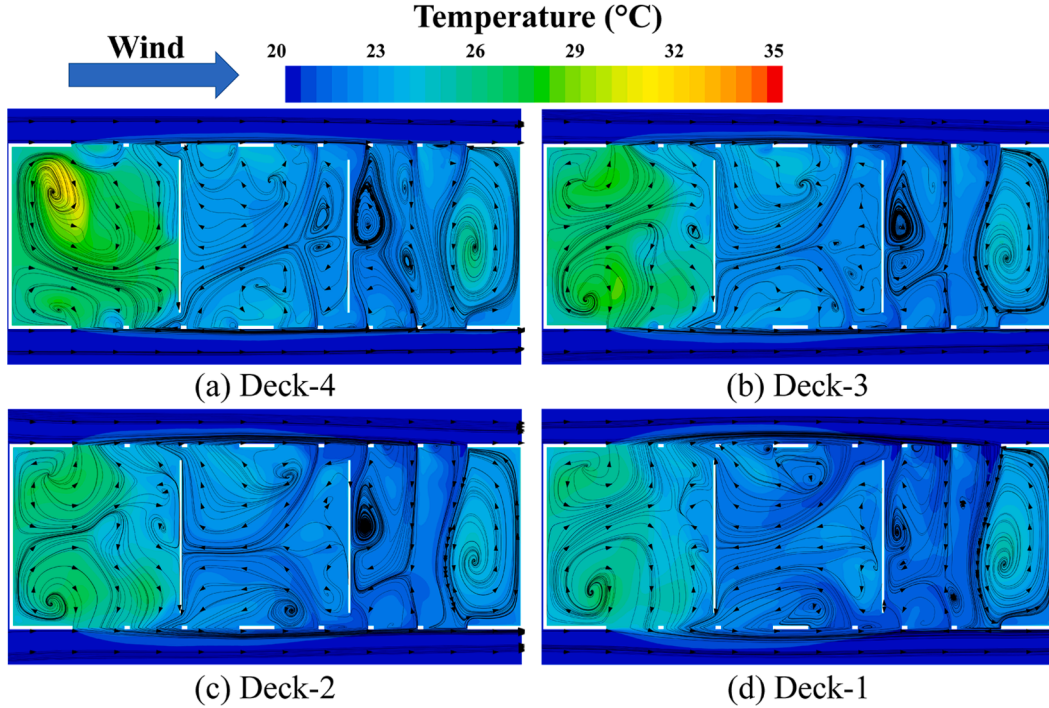


Fig. 5. Air temperature distributions covered by streamlines at horizontal planes (50 cm above floors) at (a) Deck-4, (b) Deck-3, (c) Deck-2 and (d) Deck-1 in the case of $V_{sp} = 60 \text{ km}\cdot\text{h}^{-1}$, $H = 30 \text{ cm}$, $T_{ext} = 20 \text{ }^{\circ}\text{C}$, $W_{sp} = 5 \text{ m}\cdot\text{s}^{-1}$ and $\theta = 0^{\circ}$.

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i < j} \beta_{ij} x_i x_j + \sum_{i=1}^k \beta_{ii} x_i^2 + \varepsilon \quad (13)$$

where, y is the response; x_i and x_j are the independent variables; k is the number of independent variables; ε is the error term, and β_0 , β_i , β_{ij} and β_{ii} are the regression coefficients for intercept, linear, interaction and quadratic terms, respectively.

The RSM models were established in R with `rsm` function from 'rsm' package. The RSM models underwent a reduction process through backward elimination with step function from 'MASS' package. The partial dependence plots were generated with partial function from

'pdp' package. The partial dependence plot shows the marginal effect of interested variable by setting other variables at constant values (mean values in this study).

2.3.2. Gradient boosting machine (GBM) method

GBM method produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees (Hastie et al., 2009). In GBM, the learning procedure sequentially updates models to be maximally correlated with the negative gradient of the loss function (Natekin and Knoll, 2013), and therefore provides a more accurate estimation on the response variable. The generic gradient tree-boosting algorithm for regression is provided by the following pseudo-code

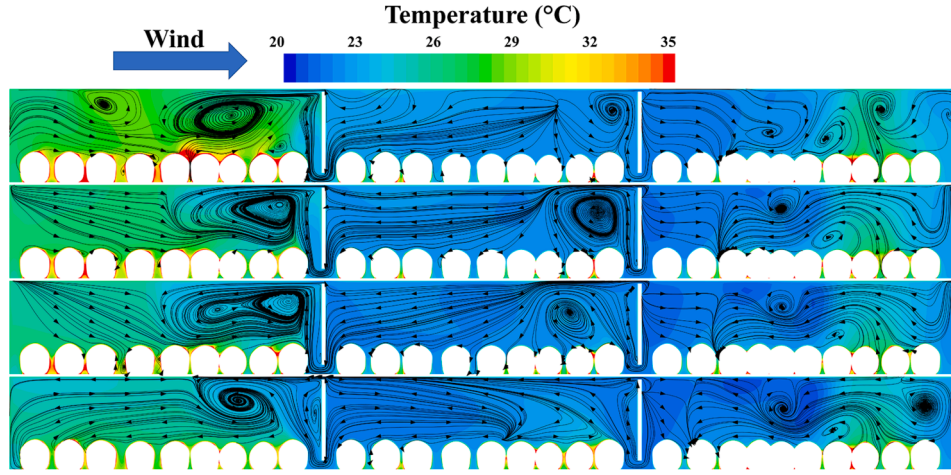


Fig. 6. Air temperature distributions covered by streamlines at central vertical planes in the case of $V_{sp} = 60 \text{ km} \cdot \text{h}^{-1}$, $H = 30 \text{ cm}$, $T_{ext} = 20^\circ \text{C}$, $W_{sp} = 5 \text{ m} \cdot \text{s}^{-1}$ and $\theta = 0^\circ$.

(Hastie et al., 2009):

1. Initialize

$$f_0(x) = \arg \min_{\gamma} \sum_{i=1}^N L(y_i, \gamma) \quad (14)$$

2. For $m = 1$ to m

- (a) For $i = 1, 2, \dots, N$ calculate residual:

$$r_{im} = - \left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \right]_{f=f_{m-1}} \quad (15)$$

- (b) Fit a regression tree to the targets r_{im} giving terminal nodes of the tree: $R_{jm}, j = 1, 2, \dots, J_m$
- (c) For $j = 1, 2, \dots, J_m$ calculate:

$$\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, f_{m-1}(x_i) + \gamma) \quad (16)$$

- (d) Update

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} I, \quad x_i \in R_{jm} \quad (17)$$

3. Get the final model

$$\hat{f}(x) = f_m(x) \quad (18)$$

where $L(x, y)$ is loss function; γ parameterizes the split variables and split points at the internal nodes, and the predictions at the terminal nodes; I is indicator function value, 1 if argument is true, and 0 otherwise.

The GBM models were established in R with train function from ‘caret’ package. Grid search method was adopted to tune hyper parameters and the final models were determined by using the smallest RMSE (root mean square error).

2.3.3. Model evaluations

The mean absolute percentage error (MAPE, %) and the RMSE were employed to evaluate the model prediction accuracy. MAPE indicates

the average magnitude of error produced by a model as expressed in Eq. (19). RMSE is the standard deviation of prediction errors, as expressed in Eq. (20). When the values of MAPE and RMSE are low, the model fits the dataset well.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - p_i}{y_i} \right| \quad (19)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - p_i)^2} \quad (20)$$

where, n , y_i and p_i are number of fitted points, actual value and predicted value, respectively.

3. Results

3.1. CFD validation

Fig. 3 shows the CFD validation results of air temperatures inside the lorry and trailer. The MAPE of using RKE2L and SKE2L were 5.3 % and 6.0 %, respectively. The CFD simulations by RKE2L and SKE2L obtained similar air temperatures in the lorry. The steady-state simulation might fail to capture the heat accumulation during unsteady-state (e.g. changes in vehicle speed) and the transient impact of exterior flow, which is a possible explanation for the large discrepancy between CFD simulations and experimental measurements in the lorry. The CFD-predicted air temperatures in the trailer, especially obtained by using RKE2L, were closer to the measured data. Since cooler exterior airflow seldom entered the trailer, the natural ventilation in the trailer was mainly driven by thermal force and the temperature distribution inside the trailer was less affected by the exterior wind condition, therefore the steady-state simulations predicted the temperature distribution inside the trailer considerably. Considering the uncertainties in defining accurate incoming wind condition, air leakage of vehicle (Norton et al., 2013) and dynamic movement of pigs, a maximum discrepancy of 4°C in CFD predictions (in the front compartment at Deck 1) by using RKE2L was acceptable. Therefore, this study used RKE2L with updating heat fluxes of pigs’ surfaces to model the air temperature distribution inside the moving vehicle.

3.2. CFD-simulated air temperature distributions

Fig. 4 shows the streamlines around the moving vehicle and airflow exchanges through the openings when wind was parallelly blown to the vehicle. Vortices were generated beside the doors of driver’s cabin and

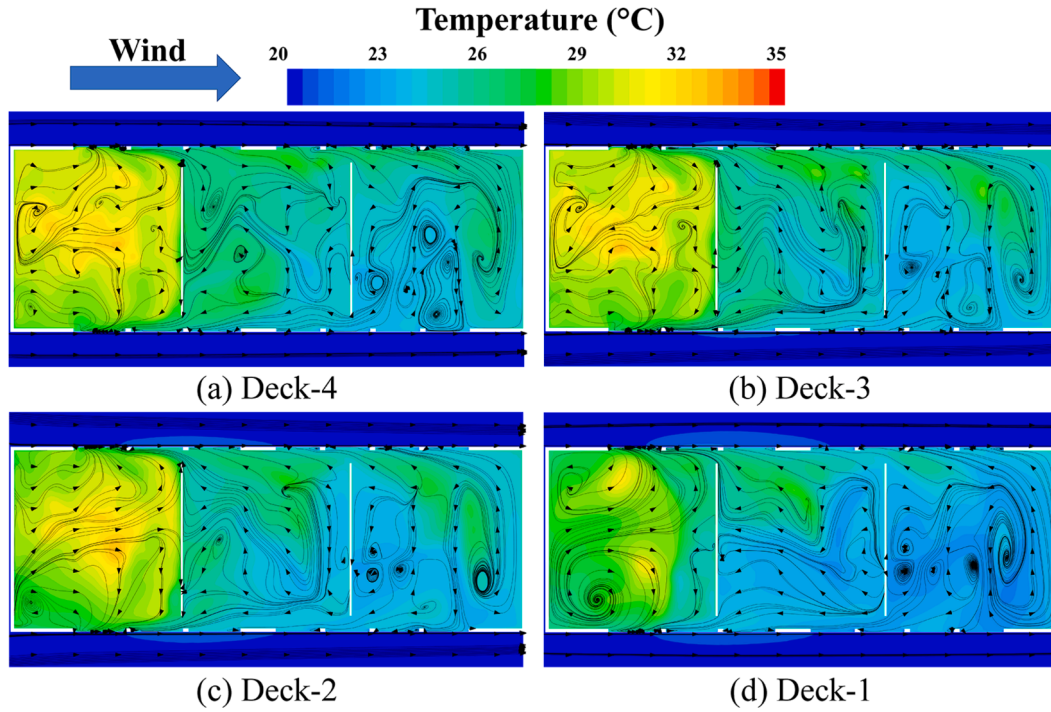


Fig. 7. Air temperature distributions covered by streamlines at horizontal planes (50 cm above floors) at (a) Deck-4, (b) Deck-3, (c) Deck-2 and (d) Deck-1 in the case of $V_{sp} = 60 \text{ km}\cdot\text{h}^{-1}$, $H = 6 \text{ cm}$, $T_{ext} = 20 \text{ }^{\circ}\text{C}$, $W_{sp} = 5 \text{ m}\cdot\text{s}^{-1}$ and $\theta = 0^{\circ}$.

thereafter airflows were separated from lorry's front part, then reattached to the lorry at the rear part.

Fig. 5 shows that the horizontal distributions of air temperature and streamlines at 50 cm above the floors of compartments at the same column (e.g., front, middle and rear) were very similar except those in the front compartment at Deck-4 when the openings were fully opened. The ventilation in the rear compartments were mainly driven by wind force with considerable cooler exterior air entering the last three openings and vortices generated near the rear walls and partition walls. Strong air mixings were observed in middle and rear compartments, see Fig. 4. However, the airflow exchanges between compartments were hindered by the adopted partitions with only partial cooler air flowed forward into the next compartments through the gaps near floors, see Figs. 6 and 8. The ventilation in front compartments was mainly driven by buoyancy force with considerable warmer interior air being purged

out from the first two openings. When opening height was decreased to 6 cm, ventilation in all compartments seemed to be dominated by buoyancy force, and the air temperatures in each compartment at both horizontal and vertical directions increased considerably, see Figs. 7 and 8.

Fig. 9 summarized the spatial distributions of air temperature at 50 cm above floor at each deck of these two presented cases. When the opening height was enlarged from 6 cm to 30 cm, with other conditions being fixed, the reduction in plane-average air temperature was up to $4.1 \text{ }^{\circ}\text{C}$ which was observed in the front compartment at Deck-2. However, large variations in air temperature within compartment were observed with both large and small opening heights, especially in the front compartments. The maximum differences between maximum and minimum value of air temperature were $9.8 \text{ }^{\circ}\text{C}$ (in the front compartment at Deck 4) and $7.5 \text{ }^{\circ}\text{C}$ (in the front compartment at Deck-2) with

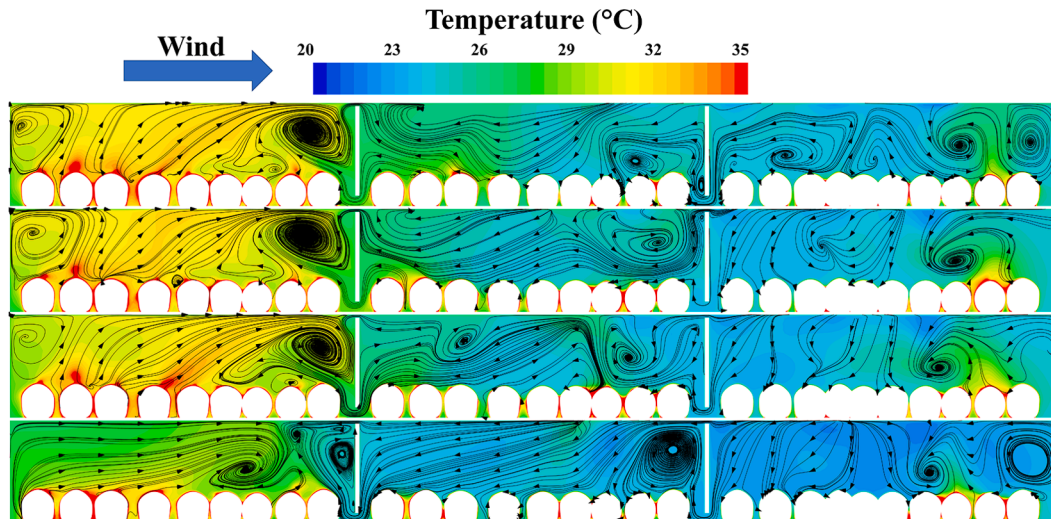


Fig. 8. Air temperature distributions covered by streamlines at central vertical plane in the case of $V_{sp} = 60 \text{ km}\cdot\text{h}^{-1}$, $H = 6 \text{ cm}$, $T_{ext} = 20 \text{ }^{\circ}\text{C}$, $W_{sp} = 5 \text{ m}\cdot\text{s}^{-1}$ and $\theta = 0^{\circ}$.

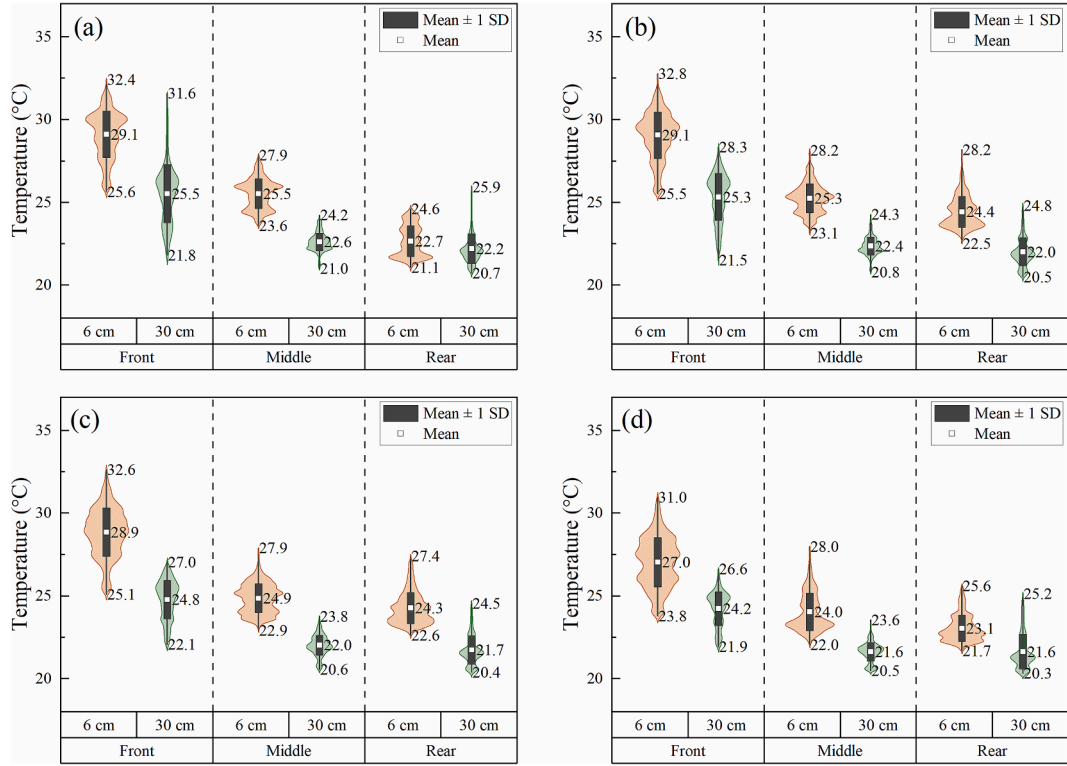


Fig. 9. Spatial distributions of air temperature at planes 50 cm above floors at (a) Deck-4, (b) Deck-3, (c) Deck-2 and (d) Deck-1 in the cases of $V_{sp} = 60 \text{ km} \cdot \text{h}^{-1}$, $T_{ext} = 20^\circ \text{C}$, $W_{sp} = 5 \text{ m} \cdot \text{s}^{-1}$, $\theta = 0^\circ$ with opening height of 6 cm and 30 cm, respectively.

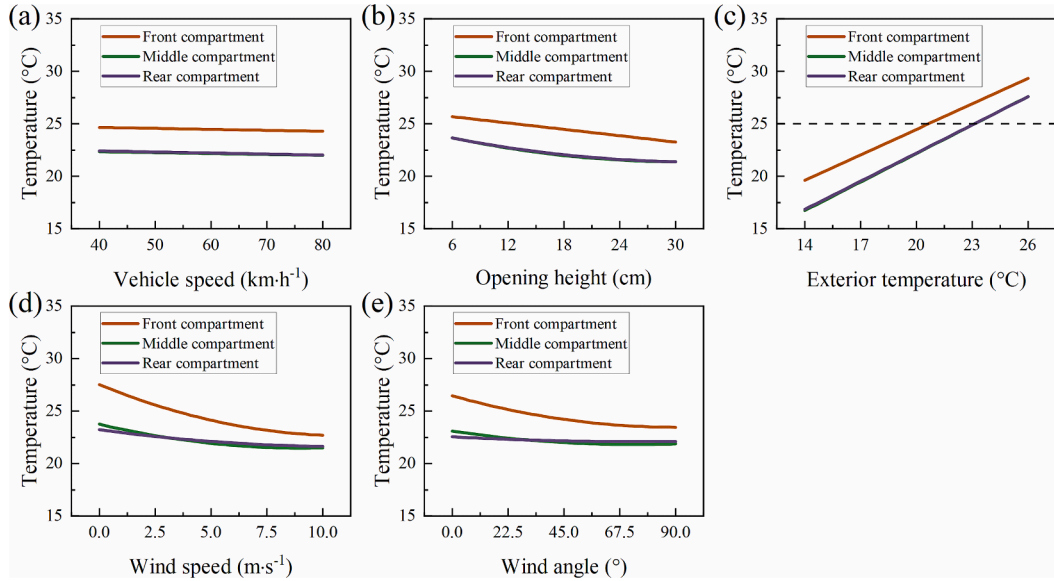


Fig. 10. Partial dependence plots with respect to the plane-average air temperatures at 50 cm above floors in the three compartments derived from the RSM models with wind angles of 0° to 90° .

opening height of 30 cm and 6 cm, respectively. The vertical variation at three different heights (40 cm, 50 cm and 60 cm) of these two presented cases were as small as negligible in terms of plane-average air temperature, the figures are not shown, which indicates that it is reasonable to set the plane-average air temperatures at 50 cm above the floors as the response variables in this study and the horizontal variation in air temperature within/between compartment(s) was more imperative to be considered.

3.3. Predictive models

3.3.1. RSM models

The final RSM models were listed in Eqs. (21) to (23). The RSM models fitted the plane-average air temperatures in middle and rear compartments well with R_{adj}^2 higher than 0.97. While the RSM models predicted the plane-average air temperature in the front compartments with slightly less accuracy ($R_{adj}^2 = 0.95$ and 0.93), especially when wind came from the back of the moving vehicle.

$$T_{\text{front}} = \begin{cases} 20.3 + 3.69 \times 10^{-4} \theta^2 + 3.93 \times 10^{-2} W_{\text{sp}}^2 - 4.9 \times 10^{-2} \theta - 1.8 W_{\text{sp}} + 0.65 T_{\text{ext}} - 0.1 H - 4.66 \times 10^{-2} V_{\text{sp}} - 3.49 \times 10^{-3} \theta \bullet W_{\text{sp}} \\ \quad + 3.18 \times 10^{-2} W_{\text{sp}} \bullet T_{\text{ext}} + 7.44 \times 10^{-3} W_{\text{sp}} \bullet V_{\text{sp}}, \theta \in [0, 90] \\ 30.8 + 1.06 \times 10^{-3} \theta^2 + 4.62 \times 10^{-2} W_{\text{sp}}^2 - 0.26 \theta - 2.25 W_{\text{sp}} + 0.78 T_{\text{ext}} - 0.11 H \\ \quad + 1.03 \times 10^{-2} \theta \bullet W_{\text{sp}} + 1.63 \times 10^{-4} \theta \bullet V_{\text{sp}}, \theta \in [90, 180] \end{cases} \quad (21)$$

$$T_{\text{middle}} = \begin{cases} 15.2 + 2.41 \times 10^{-4} \theta^2 + 2.77 \times 10^{-2} W_{\text{sp}}^2 + 3.87 \times 10^{-3} H^2 - 3.52 \times 10^{-2} \theta - 1.2 W_{\text{sp}} + 0.74 T_{\text{ext}} - 0.37 H - 3.04 \times 10^{-2} V_{\text{sp}} \\ \quad + 1.54 \times 10^{-2} W_{\text{sp}} \bullet T_{\text{ext}} + 7.28 \times 10^{-3} H \bullet W_{\text{sp}} + 5.06 \times 10^{-3} H \bullet T_{\text{ext}} + 4.39 \times 10^{-3} W_{\text{sp}} \bullet V_{\text{sp}}, \theta \in [0, 90] \\ 18.0 + 6.15 \times 10^{-4} \theta^2 + 1.67 \times 10^{-2} W_{\text{sp}}^2 + 2.46 \times 10^{-3} H^2 - 0.15 \theta - 1.08 W_{\text{sp}} \\ \quad + 0.89 T_{\text{ext}} - 0.2 H + 5.63 \times 10^{-3} \theta \bullet W_{\text{sp}} - 1.72 \times 10^{-4} \theta \bullet V_{\text{sp}}, \theta \in [90, 180] \end{cases} \quad (22)$$

$$T_{\text{rear}} = \begin{cases} 13.2 + 7.85 \times 10^{-5} \theta^2 + 1.28 \times 10^{-2} W_{\text{sp}}^2 + 3.2 \times 10^{-3} H^2 - 1.21 \times 10^{-2} \theta - 0.76 W_{\text{sp}} + 0.74 T_{\text{ext}} - 0.32 H - 3.17 \times 10^{-2} V_{\text{sp}} \\ \quad + 1.03 \times 10^{-2} W_{\text{sp}} \bullet T_{\text{ext}} + 5.59 \times 10^{-3} H \bullet T_{\text{ext}} + 4.39 \times 10^{-3} W_{\text{sp}} \bullet V_{\text{sp}}, \theta \in [0, 90] \\ 15.0 + 2.29 \times 10^{-4} \theta^2 - 0.04 \theta - 0.92 W_{\text{sp}} + 0.76 T_{\text{ext}} - 0.36 H + 6.5 \times 10^{-3} \theta \bullet W_{\text{sp}} \\ \quad + 6.42 \times 10^{-3} H \bullet T_{\text{ext}} - 4.3 \times 10^{-4} \theta \bullet V_{\text{sp}} + 1.76 \times 10^{-3} H \bullet V_{\text{sp}}, \theta \in [90, 180] \end{cases} \quad (23)$$

where T_{front} , T_{middle} and T_{rear} are the plan-average air temperature ($^{\circ}\text{C}$) at 50 cm above floors of front, middle and rear compartments, respectively.

The partial dependence plots of the five investigated factors derived from RSM models in wind angles of 0° – 90° and 90° – 180° were shown in Fig. 10 and Fig. 11, respectively. The partial dependence plots indicate that the air temperatures in the front compartments were considerably distinct from those of the other two compartments. Meanwhile, the partial dependence plots of vehicle speed, opening height and exterior temperature derived from RSM models were nearly the same for the middle and rear compartments in both wind angles of 0° – 90° and 90° – 180° . The interior air temperatures increased linearly with exterior air temperature, see Fig. 10c and Fig. 11c, when the opening height ranged from 6 cm to 30 cm. Increasing opening height, wind speed and making the wind blow normally to the vehicle (approx. 110° for front and middle compartments, 90° for rear compartment) would help to cool down the interior of lorry to some extent. The impact of vehicle

speed on interior temperatures seemed to be marginal.

3.3.2. GBM models

GBM was employed to generate three models for front, middle and rear compartments, respectively, covering the full wind angle of 0° – 180° . Meanwhile, GBM was also employed to generate a unitary model for any of the twelve compartments covering the full wind angle of 0° – 180° . Due to the complexity of the model structure, the developed GBM models were not listed in this study. Beside integrating the wind angle, the GBM models provided better prediction on the air temperature inside the front compartments than the RSM models, but improvements on prediction accuracies on the air temperatures inside middle and rear compartments were moderate, as shown in Fig. 12. The unitary GBM model also yielded satisfactory performance, as shown in Fig. 12.

The summaries of the GBM models for each of those three compartments indicated that the exterior temperature was the most

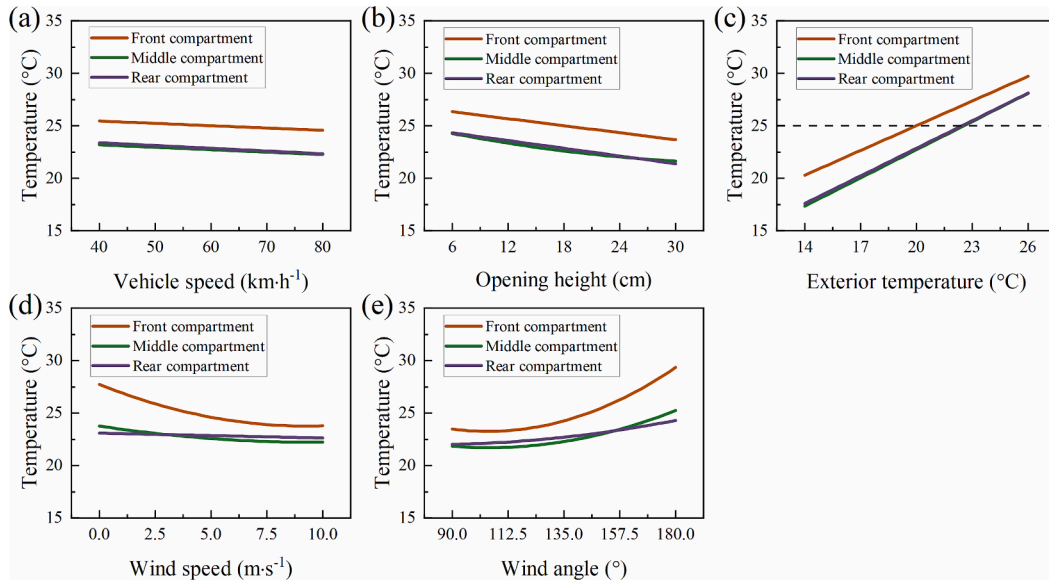


Fig. 11. Partial dependence plots with respect to the plane-average air temperatures at 50 cm above floors in the three compartments derived from the RSM models with wind angles of 90° to 180° .

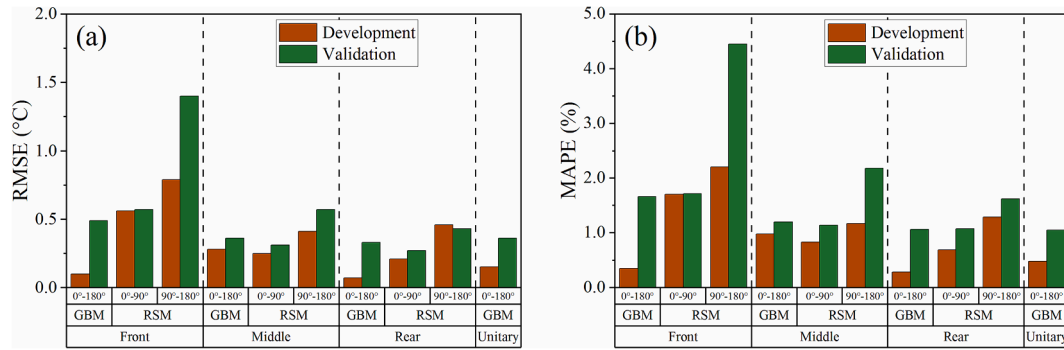


Fig. 12. Model evaluations in terms of (a) RMSE and (b) MAPE.

important influencing factor on the air temperatures inside the lorry, see Fig. 13. The relative influence of exterior temperature increased horizontally from front to rear compartments. The wind angle ranked as the second and wind speed being the third important influencing factor for front and middle compartments. The opening height ranked as the second important influencing factor for the rear compartment and fourth for front and middle compartment. The vehicle speed had the lowest influence on the air temperatures inside the moving lorry. The relative influence of each independent variable obtained by GBM models was consistent with tendencies of partial dependence plots in Fig. 10 and Fig. 11.

The partial dependency of compartment from the unitary GBM model showed that differences in air temperature between middle and rear compartment were as small as negligible, as their mean value were 22.5 °C and 22.6 °C, respectively. The front compartment, with mean value of 24.7 °C, was approx. 2.2 °C higher than the other two compartments at the same deck in term of plan-average when setting the other variable as their corresponding mean values. The partial dependency of deck from the unitary GBM model showed that the air temperature increased vertically with mean value of 23.0 °C for Deck-1, 23.2 °C for Deck-2, 23.3 °C for Deck-3 and 23.6 °C for Deck-4, respectively. It might probably be caused by the shape of cabin's rooftop which affected the separation and reattachment of exterior flows around the vehicle, see the streamlines in Fig. 4.

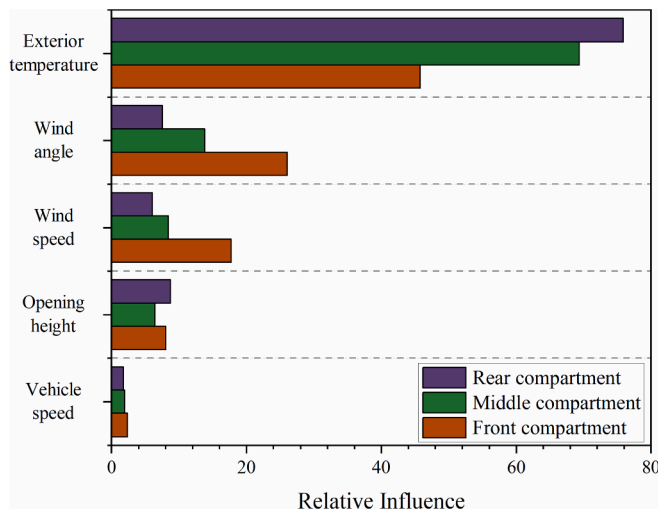


Fig. 13. Relative influence of the factors with respect to the plane-average air temperatures at 50 cm above floors in the three compartments derived from the GBM models for each of the three compartments, respectively.

4. Discussion

4.1. Vehicle design

EFSA Ahaw Panel, 2004 suggested that the opening height of naturally ventilated vehicles should be wide enough (e.g. 40 cm high). The naturally ventilated livestock vehicle modelled in this study has maximum opening height of 30 cm. The impact of opening height on interior air temperatures, shown in Fig. 10b and Fig. 11b, illustrated that enlarging opening height (with identical opening length) can gradually reduce the interior air temperature since slightly higher amount of exterior air could be supplied to the compartments. But it is also noticed that the opening height impacts the interior air temperature of compartment at different level, depending on the positions of the compartment, e.g., front, middle or rear. Following the structure of the experimental livestock vehicle, the openings was placed 40 cm above the compartment floors, far above the head of lying pigs. If the openings were further enlarged, making the bottom of the openings closer to the floors, the cooler exterior air could directly flow through pigs' skin and convection would thereafter being strengthened. Further study should be conducted to investigate the optimum design of openings (e.g., size and location) in order to improve the thermal conditions in front compartments.

The contours from Fig. 5 to Fig. 8 indicate that the interior flows were hindered by the adopted solid partitions, the thermal conditions in the front compartments are expected to be improved when the partitions were removed or being replaced with perforated one. Meanwhile, insulating the vehicle is expected to improve the thermal conditions under both warmer and colder weather conditions although this has not been examined in this study.

4.2. Placement of monitor sensors

The results indicated that the front compartments had notably higher air temperature than the other two compartments at the same deck under warmer weather, which was consistent with our experimental observations (Foldager et al., 2022). However, when being evaluated in terms of plane-average value, the differences in air temperature between front and rear compartments were less significant than the experimental data obtained from sensors installed 40 cm above the compartment floors and at the rear wall of each compartment. Fig. 9 shows that high deviation of air temperature occurred inside each compartment, especially inside the front compartments where the difference between maximum and minimum temperatures could be up to 9.8 °C. It further proves that evaluating the thermal condition inside each compartment by using the measured value from limited sensors is insufficient. Goldhawk et al (2014b) proposed that distributing ten sensors evenly throughout a horizontal plane in the compartment was more adequate to monitor the temperature distribution when transporting cattle under warmer weather. In situ experiments could be further conducted to

investigate the optimum displacement of temperature sensors for evaluating the precise temperature distribution inside each compartment follows the method proposed by Lee et al. (2019). Monitoring the physiological condition of animals and interior environmental condition around animal directly by wearable sensors (Zhang et al., 2020) could also be considered in the future research.

4.3. Application of the developed models

With a more complicated model construction, GMB models yielded slightly better prediction performances than the second-order polynomial RSM models in terms of MAPE and RMSE. However, regarding the accuracy of prediction required by industry, time consumed on tuning the models and easy interpretation on the established models, RSM models seemed to be sufficient and efficient on predicting the thermal conditions inside moving naturally ventilated livestock vehicle.

Our previous on-road experiments indicated that the built-in sensors installed on the sidewalls failed to represent the interior temperatures to a large extent compared with the sensors installed near the animal occupied zone (Herskin et al., 2022). Consequently, the driver was misinformed and failed to regulate the opening height of the shutters and activate the fans in time. As it is still a challenging for optimum displacement of monitor sensors due to high spatial variation of air temperature inside the moving transport vehicle, predictive models are thereby needed to forecast the interior temperature distribution with reasonable reliability. The developed RSM models could predict the plane-average air temperature with inputs of opening height of shutters, vehicle speed, and weather data (air temperature, wind speed and wind direction) from the nearest weather station via real-time GPS data, and hereafter inform the driver to regulate the ventilation systems in advance. For instance, the plan-average air temperature at 50 cm above the floors of the front compartments could exceed the upper threshold (25 °C) of the thermal neutral zone for weaner pigs (EFSA Ahaw Panel et al., 2022) when the input variables being their corresponding mean values in the prediction range of the models, see the black dot lines in Fig. 10c and Fig. 11c. Reminder signal will be thereby given to driver to increase the opening height or further activate the fans. Meanwhile, the real-time predicted data and recorded data from built-in sensors could be further coupled to improve the performance of the developed models.

4.4. Limitations and perspectives

In this study, the effect of stocking density was not considered. Extra models considering long journey which has lower stocking density can be examined. Further investigations on how to optimize the stocking density for improving the local thermal conditions in front compartments can be conducted. Meanwhile, the current study only focused on warmer period in Denmark. Further development of predictive models during colder period with shutters being fully closed or slightly opened are also worthy of investigation to ascertain the interior thermal conditions and air quality. The impact of solar radiation and wall insulation on the temperature distributions inside the vehicle, especially when vehicle was stationary, is not investigated. The steady-state CFD simulations are not able to capture the dynamics of air temperature induced by, e.g., dynamic movements of pigs, dynamics of thermal storage.

Appendix

The dimension of window grill was measured manually from the experimental vehicle. The height and width of window grill were 28 cm and 58 cm, respectively. The fan installed in the experimental vehicle was axial motor fan manufactured by SPAL with model number of VA11 – BP12/C – 29A, and its 3D model was downloaded from the internet. The thicknesses of window grill and fan were 5 mm and 28 mm, respectively. The geometric models of computation domains were shown in Fig. A1. The meshes were generated following the methodology described in subsection 2.1.2 with final cell number of 2.4 million and 2.6 million for the cases of window grill and fan, respectively. The inlet, outlet and other sidewalls of the domains were set as velocity inlet, pressure outlet and symmetry plane, respectively. The pressure drops between inlet and outlet under different superficial

Further study could consider these dynamics and investigate the transient thermal conditions by conducting unsteady-state CFD simulations.

5. Conclusions

This study developed predictive models for front, middle and rear compartments by employing Box-Behnken design, RSM and CFD modelling, and one model for the entire lorry by employing GBM and CFD modelling. Both methods provide acceptable predictive accuracy with high R^2_{adj} . The models developed by RSM might be more popular in truck industry since it is a polynomial relationship between predicted interior air temperature and influencing factors to be considered. Notably, opening heights impact the indoor thermal conditions of compartments at different level depending on the positions of the compartment. Thus, the maximum opening height may not be necessary to be identical for all compartments. Unevenly distributed indoor air temperature imposes challenging of installing temperature sensors to reflect the indoor thermal conditions. Further studies on locations of built-in temperature sensors are needed to utilize the predictive models in an efficient manner and regulate the openings and fans via reasonable feedback from temperature sensors. With further improvement, the predictive models are expected to assist in predicting interior air temperatures in advance and therefore regulate the openings and fans ahead to ensure the proper indoor thermal environment for transporting weaner pigs.

CRediT authorship contribution statement

Guoxing Chen: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Guoqiang Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Li Rong:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

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velocities (imposed at the inlet) were used to calculate the porous-media resistance with Eq. (11) and Eq. (12).

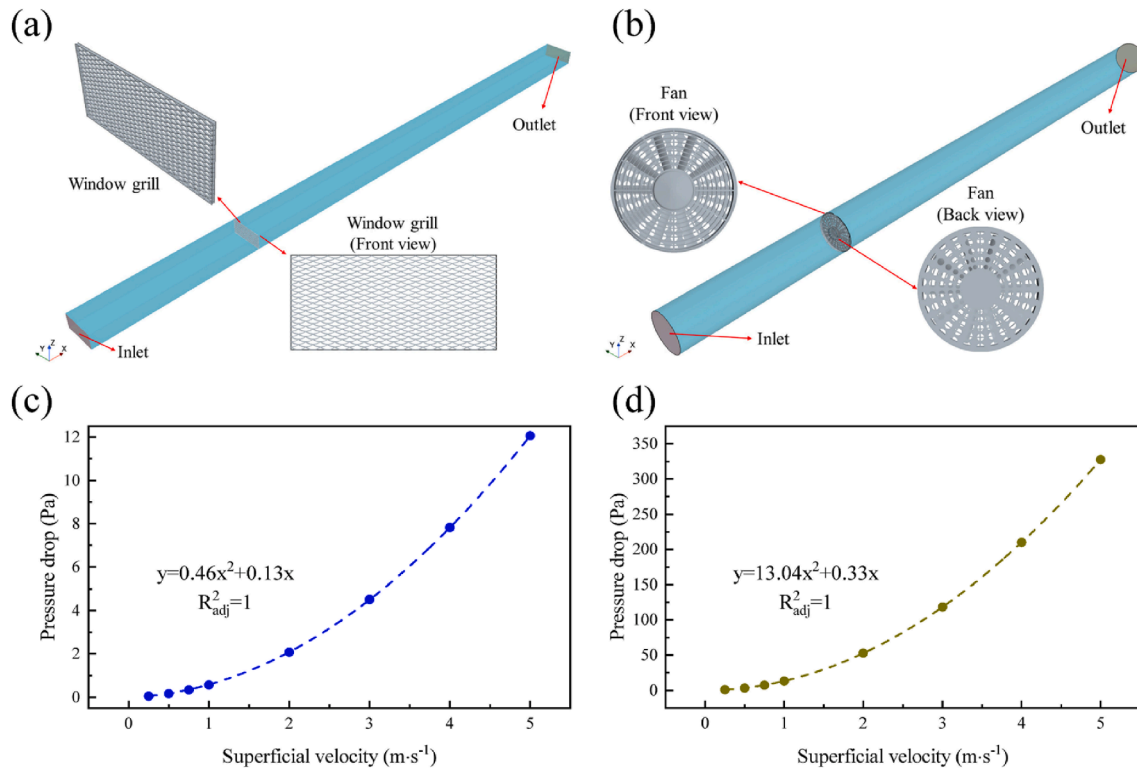


Fig. A1. Geometric models of computation domain in modelling (a) window grill and (b) fan; Pressure drops under different superficial velocities in the case of (c) window grill and (d) fan.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109591>.

Data availability

Data will be made available on request.

References

- Abdessalem, B., Hargreaves, D., Morvan, H., 2012. CFD simulations of crosswind impinging on a high speed train model. Presented at the 18th Australasian Fluid Mechanics Conference, Launceston, Australia.
- Ahmad, M.W., Mourshed, M., Rezgui, Y., 2017. Trees vs Neurons: Comparison between random forest and ANN for high-resolution prediction of building energy consumption. *Build. 147*, 77–89. <https://doi.org/10.1016/j.enbuild.2017.04.038>.
- Alawadi, S., Mera, D., Fernández-Delgado, M., Alkhabbas, F., Olsson, C.M., Davidsson, P., 2022. A comparison of machine learning algorithms for forecasting indoor temperature in smart buildings. *Energy Syst. 13*, 689–705. <https://doi.org/10.1007/s12667-020-00376-x>.
- Arulmozhi, E., Moon, B.E., Basak, J.K., Sihalath, T., Park, J., Kim, H.T., 2021. Machine Learning-Based Microclimate Model for Indoor Air Temperature and Relative Humidity Prediction in a Swine Building. *Animals 2021*, Vol. 11, Page 222 11, 222. <https://doi.org/10.3390/ANI11010222>.
- Boehmke, B., Greenwell, B.M., 2019. *Hands-on machine learning with R*. CRC Press.
- Brown, J.A., Samarakone, T.S., Crowe, T., Bergeron, R., Widowski, T., Correa, J.A., Faucitano, L., Torrey, S., Gonyou, H.W., 2011. Temperature and humidity conditions in trucks transporting pigs in two seasons in eastern and western Canada. *Trans. ASABE 54*, 2311–2318. <https://doi.org/10.13031/2013.40650>.
- CD-adapco, 2021. *Simcenter STAR-CCM+ 2021.3 User Guide*.
- Chen, G., Rong, L., Zhang, G., 2021a. Unsteady-state CFD simulations on the impacts of urban geometry on outdoor thermal comfort within idealized building arrays. *Sustain. Cities Soc. 74*, 103187. <https://doi.org/10.1016/j.scs.2021.103187>.
- Chen, G., Rong, L., Zhang, G., 2021b. Impacts of urban geometry on outdoor ventilation within idealized building arrays under unsteady diurnal cycles in summer. *Build. Environ. 206*, 108344. <https://doi.org/10.1016/j.buildenv.2021.108344>.
- Chen, G., Zhang, G., Bjerg, B., Pedersen, P., Jensen, T., Rong, L., 2024. CFD investigation on a novel pen partition-attached jet air supply for mitigating heat stress among lactating sows. *Comput. Electron. Agric. 220*, 108840. <https://doi.org/10.1016/j.compag.2024.108840>.
- Christensen, L., Gade, P.B., 1997. Design of experimental vehicle for transport of pigs and some preliminary results of environmental measurements. Danish Meat Research Institute.
- Christensen, L., Jonsson, K., 2007. Optimization of transport conditions in relation to transport mortality. Danish Meat Research Institute.
- Condotta, I.C.F.S., Brown-Brandl, T.M., Stinn, J.P., Rohrer, G.A., Davis, J.D., Silva-Miranda, K.O., 2018. Dimensions of the modern pig. *Trans. ASABE 61*, 1729–1739. <https://doi.org/10.13031/trans.12826>.
- Dobbinson, S., 2019. A study of factors that lead to stress in pigs during road transport including an examination of a novel mechanical ventilation system (Thesis). Lincoln University.
- EFSA AHAW Panel (EFSA Panel on Animal Health and Welfare), Nielsen, S., Alvarez, J., Bicout, D., Calistri, P., Canali, E., Drewe, J., Garin-Bastuji, B., Gonzales Rojas, J., Gortazar Schmidt, C., Michel, V., Miranda Chueca, M., Padalino, B., Pasquali, P., Roberts, H., Spoolder, H., Stahl, K., Velarde, A., Viltrop, A., Winckler, C., Earley, B., Edwards, S., Faucitano, L., Marti, S., Miranda de La Loma, G., Nanni Costa, L., Thomsen, P., Ashe, S., Mur, L., Van der Stede, Y., Herskin, M., 2022. Scientific opinion on the welfare of pigs during transport. *EFSA J. 20*, e07445.
- EFSA AHAW Panel (EFSA Panel on Animal Health and Welfare), 2004. Opinion of the Scientific Panel on Animal Health and Welfare on a request from the Commission related to the Standards for the microclimate inside animal road transport vehicles. *EFSA Journal 2*, 1–25. <https://doi.org/10.2903/j.efsa.2004.122>.
- Foldager, L., Kaiser, M., Chen, G., Jensen, L.D., Kristensen, J.Kr., Kobek-Kjeldager, C., Thodberg, K., Li, R., Herskin, M.S., 2022. Deck height during transport of weaners - piglet height and microclimatic conditions inside the vehicle, in: Symposium i Anvendt Statistik 2022-2. Presented at the Det 43. Symposium i Anvendt Statistik, København.
- Gautam, K.R., Zhang, G., Landwehr, N., Adolphs, J., 2021. Machine learning for improvement of thermal conditions inside a hybrid ventilated animal building. *Comput. Electron. Agric. 187*, 106259. <https://doi.org/10.1016/j.compag.2021.106259>.
- Gerritsen, M.A., Hindle, V.A., Steinkamp, K., Reimert, H.G.M., van der Werf, J.T.N., Marahrens, M., 2013. The effect of reduced loading density on pig welfare during long distance transport. *Animal 7*, 1849–1857. <https://doi.org/10.1017/S1751731113001523>.

- Gilkeson, C.A., Thompson, H.M., Wilson, M.C.T., Gaskell, P.H., 2016. Quantifying passive ventilation within small livestock trailers using computational fluid dynamics. *Comput. Electron. Agric.* 124, 84–99. <https://doi.org/10.1016/j.compag.2016.03.028>.
- Goldhawk, C., Crowe, T., Janzen, E., González, L.A., Kastelic, J., Pajor, E., Schwartzkopf-Genswein, K.S., 2014a. Trailer microclimate during commercial transportation of feeder cattle and relationship to indicators of cattle welfare. *J. Animal Sci.* 92, 5155–5165. <https://doi.org/10.2527/jas.2014-7964>.
- Goldhawk, C., Kastelic, J., Schwartzkopf-Genswein, K., Goldhawk, C., Janzen, E., Pajor, E., Crowe, T., González, L.A., 2014b. Comparison of eight logger layouts for monitoring animal-level temperature and humidity during commercial feeder cattle transport. *J. Anim. Sci.* 92, 4161–4171. <https://doi.org/10.2527/JAS.2013-7432>.
- Haley, C., Dewey, C.E., Widowski, T., Friendship, R., 2008. Association between in-transit loss, internal trailer temperature, and distance traveled by Ontario market hogs. *Can. J. Vet. Res.* 72, 385–389.
- Harmon, J.D., Hoff, S.J., Baas, T.J., Zhao, Y., Xin, H., Follett, L.R., 2017. Evaluation of conditions during weaned pig transport. *Appl. Eng. Agric.* 33, 901–912. <https://doi.org/10.13031/aea.12367>.
- Hastie, T., Tibshirani, R., Friedman, J., 2009. Boosting and Additive Trees, in: Hastie, T., Tibshirani, R., Friedman, J. (Eds.), *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Springer Series in Statistics. Springer, New York, NY, pp. 337–387. https://doi.org/10.1007/978-0-387-84858-7_10.
- Herskin, M.S., Kobek-Kjeldager, C., Kaiser, M.O., Thodberg, K., Jensen, L.D., Chen, G., Zhang, G., Rong, L., Foldager, L., 2022. Krav til indvendig højde ved transport af smågrise (rådgivningsnotat fra DCA – nationalt center for fødevarer og jordbrug). Aarhus Universitet.
- Jayamaha, S.E.G., Wijesundera, N.E., Chou, S.K., 1996. Measurement of the heat transfer coefficient for walls. *Build. Environ.* 31, 399–407. [https://doi.org/10.1016/0360-1323\(96\)00014-5](https://doi.org/10.1016/0360-1323(96)00014-5).
- Kephart, K.B., Harper, M.T., Raines, C.R., 2010. Observations of market pigs following transport to a packing plant. *J. Anim. Sci.* 88, 2199–2203. <https://doi.org/10.2527/jas.2009-2440>.
- Khuri, A.I., Mukhopadhyay, S., 2010. Response surface methodology. *WIREs Comput. Stat.* 2, 128–149. <https://doi.org/10.1002/wics.73>.
- Lambooy, E., 1988. Road transport of pigs over a long distance: some aspects of behaviour, temperature and humidity during transport and some effects of the last two factors. *Anim. Sci.* 46, 257–263. <https://doi.org/10.1017/S000335610004232X>.
- Lee, S., Lee, I., Yeo, U., Kim, R., Kim, J., 2019. Optimal sensor placement for monitoring and controlling greenhouse internal environments. *Biosyst. Eng.* 188, 190–206. <https://doi.org/10.1016/j.biosystemseng.2019.10.005>.
- Lee, S.-Y., Lee, I.-B., Yeo, U.-H., Kim, J.-G., Kim, R.-W., 2022. Machine learning approach to predict air temperature and relative humidity inside mechanically and naturally ventilated duck houses: Application of recurrent neural network. *Agriculture* 12, 318. <https://doi.org/10.3390/AGRICULTURE12030318>.
- Li, H., Rong, L., Zhang, G., 2018a. Numerical study on the convective heat transfer of fattening pig in groups in a mechanical ventilated pig house. *Computers and electronics in agriculture, SI: CFD in Agri. Bio. Eng.* 149, 90–100. <https://doi.org/10.1016/j.compag.2017.08.013>.
- Ma, H., Tu, Y., Yang, X., Yang, Z., Liang, C., 2022. Influence of tunnel ventilation on the indoor thermal environment of a poultry building in winter. *Build. Environ.* 223, 109448. <https://doi.org/10.1016/j.buildenv.2022.109448>.
- Ministry of Food, Agriculture and Fisheries, 2020. Bekendtgørelse om beskyttelse af dyr under transport.
- Mostafa, E., Lee, I.-B., Song, S.-H., Kwon, K.-S., Seo, I.-H., Hong, S.-W., Hwang, H.-S., Bitog, J.P., Han, H.-T., 2012. Computational fluid dynamics simulation of air temperature distribution inside broiler building fitted with duct ventilation system. *Biosyst. Eng.* 112, 293–303. <https://doi.org/10.1016/j.biosystemseng.2012.05.001>.
- Natekin, A., Knoll, A., 2013. Gradient boosting machines, a tutorial. *Front. Neurobot.* 7.
- Norton, T., Grant, J., Fallon, R., Sun, D.-W., 2010. Assessing the ventilation performance of a naturally ventilated livestock building with different eave opening conditions. *Comput. Electron. Agric.* 71, 7–21. <https://doi.org/10.1016/j.compag.2009.11.003>.
- Norton, T., Kettlewell, P., Mitchell, M., 2013. A computational analysis of a fully-stocked dual-mode ventilated livestock vehicle during ferry transportation. *Comput. Electron. Agric.* 93, 217–228. <https://doi.org/10.1016/j.compag.2013.02.005>.
- Papadopoulos, S., Azar, E., Woon, W.-L., Kontokosta, C.E., 2018. Evaluation of tree-based ensemble learning algorithms for building energy performance estimation. *J. Build. Perform. Simul.* 11, 322–332. <https://doi.org/10.1080/19401493.2017.1354919>.
- Pedersen, S., Sällvik, K., 2002. Climatization of Animal Houses: Heat and moisture production at animal and house levels. Research Centre Bygholm, Danish Institute of Agricultural Sciences, Horsens, Denmark.
- Qavidel Fard, Z., Zomorodian, Z.S., Korsavi, S.S., 2022. Application of machine learning in thermal comfort studies: A review of methods, performance and challenges. *Energ. Build.* 256, 111771. <https://doi.org/10.1016/j.enbuild.2021.111771>.
- R Core Team, 2021. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Randall, J.M., Patel, R., 1994. Thermally induced ventilation of livestock transporters. *J. Agric. Eng. Res.* 57, 99–107. <https://doi.org/10.1006/jaer.1994.1009>.
- Rioja-Lang, F.C., Brown, J.A., Brockhoff, E.J., Faucitano, L., 2019. A review of swine transportation research on priority welfare issues: A Canadian perspective. *Front. Veterin. Sci.* 6.
- Rong, L., Liu, D., Pedersen, E.F., Zhang, G., 2015. The effect of wind speed and direction and surrounding maize on hybrid ventilation in a dairy cow building in Denmark. *Energ. Build.* 86, 25–34. <https://doi.org/10.1016/j.enbuild.2014.10.016>.
- Shen, X., Zhang, G., Bjerg, B., 2012. Investigation of response surface methodology for modelling ventilation rate of a naturally ventilated building. *Build. Environ.* 54, 174–185. <https://doi.org/10.1016/j.buildenv.2012.02.009>.
- Shen, X., Zhang, G., Bjerg, B., 2013. Assessments of experimental designs in response surface modelling process: Estimating ventilation rate in naturally ventilated livestock buildings. *Energ. Build.* 62, 570–580. <https://doi.org/10.1016/j.enbuild.2013.03.038>.
- Sutherland, M.A., McDonald, A., McGlone, J.J., 2009. Effects of variations in the environment, length of journey and type of trailer on the mortality and morbidity of pigs being transported to slaughter. *Vet. Rec.* 165, 13–18. <https://doi.org/10.1136/vetrec.165.1.13>.
- Thodberg, K., Foldager, L., Fogsgaard, K.K., Gaillard, C., Herskin, M.S., 2022. Temperature conditions during commercial transportation of cull sows to slaughter. *Comput. Electron. Agric.* 192, 106626. <https://doi.org/10.1016/j.compag.2021.106626>.
- Touzani, S., Granderson, J., Fernandes, S., 2018. Gradient boosting machine for modeling the energy consumption of commercial buildings. *Energ. Build.* 158, 1533–1543. <https://doi.org/10.1016/j.enbuild.2017.11.039>.
- Warriss, P.D., 1998. Choosing appropriate space allowances for slaughter pigs transported by road: A review. *Vet. Rec.* 142, 449–454. <https://doi.org/10.1136/vr.142.17.449>.
- Xiong, Y., Green, A., Gates, R.S., 2015. Characteristics of trailer thermal environment during commercial swine transport managed under U.S. Industry Guidelines. *Animals* 5, 226–244. <https://doi.org/10.3390/ANI5020226>.
- Yi, Q., Zhang, G., Amon, B., Hempel, S., Janke, D., Saha, C.K., Amon, T., 2021. Modelling air change rate of naturally ventilated dairy buildings using response surface methodology and numerical simulation. *Build. Simul.* 14, 827–839. <https://doi.org/10.1007/s12273-020-0697-z>.
- Zhang, M., Feng, H., Luo, H., Li, Z., Zhang, X., 2020. Comfort and health evaluation of live mutton sheep during the transportation based on wearable multi-sensor system. *Comput. Electron. Agric.* 176, 105632. <https://doi.org/10.1016/j.compag.2020.105632>.